

Education

- 2021–? **PhD Statistics** *Carnegie Mellon University*. Pittsburgh, PA
Advised by Aaditya Ramdas
- 2019–2020 **MS Machine Learning** GPA: 4.0/4.3
- 2015–2019 **BS Computer Science** GPA: 3.88/4.0

Industry

- June-Aug. **Two Sigma**, *Quantitative Research Intern*, New York, NY
2025 Research on the systematic macro team.
- Feb-May. **Netflix**, *ML Research Intern*, New York, NY
2025 Team: *Machine Learning Inference Research*. Mentor: Michael Lindon. Manager: Nathan Kallus. Applying multiple testing and e-values to improving the A/B testing engine.
- May-Aug. **Microsoft Research**, *Research Intern*, Redmond, WA
2023 Team: *Reinforcement Learning*. Mentor: Paul Mineiro. I developed an anytime-valid method that uses active learning for calibrating the risk of black-box machine learning models [1].
- Mar.-May **Growthbook**, *Consultant*, Remote
2023 I consulted on a project for implementing safe anytime-valid inference (SAVI) methods into Growthbook's A/B testing engine. Documentation is linked [here](#) and the open source implementation is [here](#).
- June-Aug. **Twitter**, *Engineering Intern*, Remote
2022 Team: *Experimentation Data Science*. Mentors: Luke Sonnet, Umashanthi Pavalanathan. Manager: Brent Cohn. I analyzed use of safe-anytime valid inference (SAVI) methods for A/B testing.
- May-Aug. **CTRL-labs**, *Science Intern*, New York
2018 Now part of Facebook Reality Labs. I developed state-of-the-art LSTM ensemble model that models hand movement from electromyography (EMG) signals in TensorFlow.
Built parser for constructing acyclic graph pipeline for preprocessing real time EMG signals.
- May-Aug. **Bloomberg**, *Software Engineering Intern*, New York
2017 I worked on the Message Infrastructure team, where I imported RapidCheck, a Haskell QuickCheck inspired testing framework, into the Bloomberg C++ environment.
- May-Aug. **PicMonkey**, *Software Engineering Intern*, Seattle
2016 I helped build the user interface and photo editing features for the launch of the mobile photo editor app.

Languages/Frameworks

Python, Pandas, Numpy *Frequently use*
R, C, Rust, PyTorch *Familiar*

Publications

- Active, anytime-valid risk controlling prediction sets
Z. Xu, N. Karampatziakis, and P. Mineiro *NeurIPS 2024*
- Post-selection inference for e-value based confidence intervals
Z. Xu, R. Wang, and A. Ramdas *Electronic Journal of Statistics 2024*
Runner-up Poster Prize @ MCP 2022

3. Foundations of testing for finite-sample causal discovery
T. Yan, **Z. Xu**, and Z. C. Lipton ICML 2024
4. Online multiple testing with e-values
Z. Xu and A. Ramdas AISTATS 2024
5. Risk-limiting financial audits via weighted sampling without replacement
S. Shekhar, **Z. Xu**, Z. C. Lipton, P. J. Liang, and A. Ramdas UAI 2023
6. Memory bounds for the experts problem
V. Srinivas, D. P. Woodruff, **Z. Xu**, and S. Zhou STOC 2022
7. A unified framework for bandit multiple testing
Z. Xu, R. Wang, and A. Ramdas NeurIPS 2021
8. Dynamic algorithms for online multiple testing
Z. Xu and A. Ramdas Math. and Sci. ML 2021
9. Class-weighted classification: Trade-offs and robust approaches
Z. Xu, C. Dan, J. Khim, and P. Ravikumar ICML 2020
10. Strategy and policy learning for non-task-oriented conversational systems
Z. Yu, **Z. Xu**, A. W. Black, and A. Rudnicky SIGDIAL 2016
11. Chatbot evaluation and database expansion via crowdsourcing
Z. Yu, **Z. Xu**, A. W. Black, and A. Rudnicky RE-WOCHAT workshop of LREC 2016

Preprints

12. Bringing closure to false discovery rate control: A general principle for multiple testing
Z. Xu, A. Solari, L. Fischer, R. de Heide, A. Ramdas, and J. Goeman 2025
13. Active multiple testing with proxy p-values and e-values
Z. Xu, C. Wang, L. Wasserman, K. Roeder, and A. Ramdas 2025
14. An online generalization of the (e-)Benjamini-Hochberg procedure
L. Fischer, **Z. Xu**, and A. Ramdas 2024
15. More powerful multiple testing under dependence via randomization
Z. Xu and A. Ramdas 2023

Projects

- Oct. **Real Estate Auditing**, *Carnegie Mellon University*
- 2021-Dec. I am providing statistical help (e.g. data analysis, writing expert reports, etc.) for a lawsuit against Allegheny County concerning their practices for computing the assessed values (and consequently property taxes) of newly purchased homes. This was in collaboration with Barbara Stern, John Silvestri, Esq., and Prof. Aaditya Ramdas. Recent news coverage of the case is linked [here](#).
- 2022

Talks

- July 2025 **13th International Conference on Multiple Comparison Procedures**
Active multiple testing with proxy p-values and e-values
- July 2025 **Workshop on Game-theoretic Statistics and Sequential, Anytime-Valid Inference (BIRS)**
Bringing closure to FDR control: a general principle for multiple testing

- May 2025 **International Seminar on Selective Inference**
Bringing closure to FDR control with a uniform improvement of the eBH procedure
- Apr. 2025 **DeGroot Student Research Workshop (Carnegie Mellon University)**
Active multiple testing with proxy p-values and e-values
- Jul. 2022 **Twitter ML Modeling Seminar**
Valid inference under S^3 bias for A/B testing
- Jun. 2022 **Safe, Anytime-Valid Inference (SAVI) and Game-theoretic Statistics Workshop**
Post-selection inference for e-value based confidence intervals
- Mar. 2022 **International Seminar on Selective Inference**
Post-selection inference for e-value based confidence intervals
- Nov. 2021 **Waterloo Student Conference in Statistics, Actuarial Science and Finance**
A unified framework for bandit multiple testing
- Sep. 2021 **Workshop on Current and Future Trends in Multiple Hypothesis Testing**
Dynamic algorithms for online multiple testing

Teaching

Teaching Assistant

- 36–402: Advanced Methods for Data Analysis (Spring 2023, 2024)
- 36–750: Statistical Computing (Fall 2023)
- 36–650: Statistical Computing (Fall 2021)
- 15–251: Great Theoretical Ideas in Computer Science (Fall 2017, Spring 2018, Fall 2018)
- 15–150: Introduction to Functional Programming (Fall 2016, Spring 2017)

Service

- 2024–2025 Mentorship Program in Stat&DS organizer + mentor
- 2022–2023 CMU StatML Reading Group (SMLRG) organizer
- 2020 SCS Master's Advisory Committee
- 2020 MLD Master's Admissions Committee

Reviewing

- 2025 AISTATS, ICML, Statistical Methods in Medical Research, Annals of Statistics, TMLR, JASA
- 2024 NeurIPS, Biometrika, Statistica Neerlandica
- 2023 STOC, Electronic Journal of Statistics, New England Journal of Statistics in Data Science
- 2022 Mathematical and Scientific Machine Learning
- 2021 AISTATS